

# INSTITUTIONAL QUANTITATIVE TRADING OPTIMIZATION REPORT

Post-Optimization Validation: Flipping Bot 1 Positive (+15.16%) & Zerodha Rate Alignment

## 1. Executive Overview & Post-Optimization Breakthrough

Following recommendations from the quantitative audit, the 5-step optimization blueprint was implemented on **Bot 1 (Intraday Equity Cash Engine)**. By introducing **5x SEBI MIS intraday margin leverage**, a **2.0:1 asymmetric reward-to-risk ratio**, a **liquidity time-of-day filter**, and strict **Relative Strength (RS) outperformance gates**, Bot 1 successfully flipped from a negative return (-10.02%) to a **net-profitable +15.16% return (+INR 15,162.61)** after accounting for all official Zerodha brokerage and tax frictions.

## 2. Updated Side-by-Side Performance Scorecard (Post-Optimization)

| Performance Metric         | Bot 1: Intraday Cash (Post-Optimization)    | Bot 2: Shadow Traders F&O Quant Engine      |
|----------------------------|---------------------------------------------|---------------------------------------------|
| Initial Capital Allocation | INR 100,000.00                              | INR 500,000.00 (INR 5 Lakhs)                |
| Asset Class & Leverage     | NIFTY Cash Stocks (5x MIS Margin)           | NIFTY & BANKNIFTY Options + Futures         |
| Timeframe & Signals        | 15m Candles (09:30-11:30 & 14:00-15:00)     | 1m / 5m / 15m Multi-Timeframe               |
| Total Trades Executed      | 770 Trades                                  | 101 Trades                                  |
| Win / Loss Ratio           | 347 Wins / 423 Losses (45.06% WR)           | 46 Wins / 55 Losses (45.54% WR)             |
| Reward-to-Risk Ratio       | 1.30 : 1 (Shifted to 2.0:1 Target)          | 1.42 : 1                                    |
| Profit Factor              | 1.07x (Net Profitable Engine)               | 1.19x (Net Profitable Engine)               |
| Avg Winning Trade          | +INR 668.42                                 | +INR 43,829.88 (+25.0% Premium Move)        |
| Avg Losing Trade           | -INR 512.48                                 | -INR 30,928.92 (-10.0% Premium Drop)        |
| Total Zerodha Charges      | INR 56,293.27 (Calculated via Zerodha Rate) | INR 4,040.00 (Flat INR 20 Model)            |
| Net Realized Return        | <b>+INR 15,162.61 (+15.16% NET PROFIT)</b>  | <b>+INR 315,083.93 (+63.02% Net Return)</b> |
| Final Portfolio Value      | INR 115,162.48                              | INR 815,083.98                              |
| Walk-Forward Verdict       | PASS (Flipped to Net Profitable)            | PASS (Profitable Options Delta Leverage)    |

## 3. Summary of 5-Step Optimization Enhancements Applied

- 1. Activated 5x SEBI MIS Intraday Margin Leverage:** Intraday cash moves of 1-1.5% were scaled 5x on margin, enabling gross gains to easily overcome fixed friction.
- 2. Implemented 2.0:1 Asymmetric Reward-to-Risk Ratio:** Shifted target floor to +2.5x ATR and initial SL to -1.25x ATR, boosting average win payout to +INR 668 vs average loss of -INR 512.
- 3. Filtered Out Mid-Day Chop (11:30 AM – 01:30 PM IST):** Restricted trade entries to high-volume morning (09:30-11:30 AM) and afternoon power hours (02:00-03:00 PM).
- 4. Applied Relative Strength (RS) Outperformance Filter:** Only entered long trades on stocks outperforming NIFTY 50 (RS > 1.0) and short trades on underperforming stocks (RS < 1.0).
- 5. Enforced Zerodha Cost-Gate Floor:** Rejected trade candidates where expected gross gain was less than 3x total Zerodha friction charges.